

# BCA(H)<sup>1</sup> Seventh Semester Syllabus<sup>2</sup>

## Introduction to Data Science: BCAC701A

| MODULES | TOPICS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | MARKS |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1       | <b>Introduction To Statistics for Data Science:</b> <ul style="list-style-type: none"><li>• Role Of Statistics in Data Science</li><li>• Types Of Data and Variables</li><li>• Population Vs Sample</li><li>• Data Collection and Classification</li></ul>                                                                                                                                                                                                                                                                                                        | 10    |
| 2       | <b>Measures Of Central Tendency</b> <ul style="list-style-type: none"><li>• Mean, Median, Mode</li><li>• Geometric Mean, Harmonic Mean</li><li>• Applications And Simple Problems</li></ul>                                                                                                                                                                                                                                                                                                                                                                       | 15    |
| 3       | <b>Measures Of Dispersion</b> <ul style="list-style-type: none"><li>• Range, Quartile Deviation</li><li>• Mean Deviation, Standard Deviation</li><li>• Variance, Coefficient of Variation</li></ul>                                                                                                                                                                                                                                                                                                                                                               | 15    |
| 4       | <b>Correlation And Regression</b> <ul style="list-style-type: none"><li>• Karl Pearson Correlation</li><li>• Spearman Rank Correlation</li><li>• Regression Lines (X On Y, Y On X)</li><li>• Interpretation</li></ul>                                                                                                                                                                                                                                                                                                                                             | 15    |
| 5       | <b>Probability And Distributions</b> <ul style="list-style-type: none"><li>• Sample Space; Events; Types of Events</li><li>• Axiomatic Approach to Probability</li><li>• Problems Based on Permutation and Combination</li><li>• Addition Theorem</li><li>• Conditional Probability</li><li>• Multiplication Theorem</li><li>• Independent Events</li><li>• Total Probability</li><li>• Bayes' Theorem</li><li>• Binomial Distribution and Its Use</li><li>• Poisson Distribution and Its Use</li><li>• Normal Distribution (<i>Basic Idea and Use</i>)</li></ul> | 15    |
| 6       | <b>Basic Data Science Methods</b> <ul style="list-style-type: none"><li>• Introduction To Classification</li><li>• Decision Tree</li><li>• Clustering</li><li>• Simple Examples and Applications</li></ul>                                                                                                                                                                                                                                                                                                                                                        | 30    |

<sup>1</sup> From 7<sup>th</sup> Semester it becomes **BCA HONOURS**

<sup>2</sup> Link to the Original Syllabus: [https://makautexam.net/aicte\\_details/Syllabus4/BCA/sem723.pdf](https://makautexam.net/aicte_details/Syllabus4/BCA/sem723.pdf)

# Data Science Lab with Python: BCAC791A

| MODULES | TOPICS                                                                                                                                                                                                                          |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | <b>Python Fundamentals for Data Science:</b> <ul style="list-style-type: none"><li>• Variables, Data Types</li><li>• Lists, Tuples</li><li>• Dictionaries</li><li>• Loops</li><li>• Functions</li></ul>                         |
| 2       | <b>NumPy For Numerical Operations:</b> <ul style="list-style-type: none"><li>• <b>Array:</b> Creation, Indexing, Slicing</li><li>• Mathematical Operations on Arrays</li></ul>                                                  |
| 3       | <b>Pandas For Data Handling:</b> <ul style="list-style-type: none"><li>• Series And DataFrame</li><li>• Importing CSV Files</li><li>• Data Selection, Filtering</li></ul>                                                       |
| 4       | <b>Data Cleaning and Preparation</b> <ul style="list-style-type: none"><li>• Handling Missing Values</li><li>• Removing Duplicates</li><li>• Data Transformation</li></ul>                                                      |
| 5       | <b>Exploratory Data Analysis and Visualization</b> <ul style="list-style-type: none"><li>• Descriptive Statistics</li><li>• Summary Measures</li><li>• <b>Using Matplotlib:</b> Line, Bar Chart, Histogram, Pie Chart</li></ul> |
| 6       | <b>Statistical Measures and Probability Using Python</b> <ul style="list-style-type: none"><li>• Mean, Median, Mode</li><li>• Standard Deviation</li><li>• Variance</li><li>• Simple Probability Examples</li></ul>             |
| 7       | <b>Basic Data Science Techniques:</b> <ul style="list-style-type: none"><li>• <b>Implementation Of:</b> Classification, Decision Tree, Clustering on Simple Datasets</li><li>• Mini Case Study</li></ul>                        |

## Suggested Reading:

1. **Data Science and Business Analytics** — S. P. Rajagopalan, Publisher: Wiley India
2. **Statistics for Data Scientists: 50 Essential Concepts** — Peter Bruce & Andrew Bruce, Publisher: O'Reilly (Indian edition available)
3. **Fundamentals of Statistics** — S. C. Gupta & V. K. Kapoor, Publisher: Sultan Chand & Sons
4. **Data Science: Introductory and Advanced Topics** — N. G. Das, Publisher: McGraw Hill Education India
5. **Machine Learning & Data Science – An Introduction to Statistical Learning Methods with R and Python** — Daniel D. Gutierrez, Publisher: CRC Press (Indian edition)
6. **Python for Data Science** — Dr. Basant Agarwal & Dr. Sumit Chauhan, Publisher: Oxford University Press India

# Introduction to Machine Learning: BCAC701B

| MODULE | TOPICS                                                                                                                                                                                                                                                                                                                                                | MARKS |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1      | <b>Introduction to ML &amp; Data Preprocessing:</b> <ul style="list-style-type: none"><li>• Meaning and scope of ML</li><li>• Types of learning</li><li>• Applications</li><li>• ML workflow</li><li>• Data types</li><li>• Preprocessing</li><li>• Handling missing values</li><li>• Feature selection</li><li>• Training and testing data</li></ul> | 20    |
| 2      | <b>Regression Techniques:</b> <ul style="list-style-type: none"><li>• Simple linear regression</li><li>• Multiple regression (<i>basic idea</i>)</li><li>• Model prediction</li><li>• Evaluation using MAE, MSE, RMSE</li></ul>                                                                                                                       | 20    |
| 3      | <b>Classification Techniques:</b> <ul style="list-style-type: none"><li>• Logistic Regression</li><li>• KNN</li><li>• Naïve Bayes</li><li>• Confusion matrix</li><li>• Accuracy, precision, recall</li></ul>                                                                                                                                          | 20    |
| 4      | <b>Clustering and Decision Tree:</b> <ul style="list-style-type: none"><li>• K-Means clustering</li><li>• Centroid concept</li><li>• Decision tree structure</li><li>• Entropy and information gain</li><li>• Tree construction</li></ul>                                                                                                             | 20    |
| 5      | <b>Model Evaluation and Applications:</b> <ul style="list-style-type: none"><li>• Model validation</li><li>• Overfitting and underfitting</li><li>• Cross-validation (<i>basic idea</i>)</li><li>• ML applications</li><li>• Case study</li></ul>                                                                                                     | 20    |

# Machine Learning Lab: BCAC791B

| MODULE | TOPIC                                                                                                                                                                                                                               |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | <b>Python Environment and Data Handling</b> <ul style="list-style-type: none"><li>• Working In Jupyter/IDE</li><li>• Using NumPy and Pandas</li><li>• Importing CSV Datasets</li><li>• Basic Data Operations</li></ul>              |
| 2      | <b>Data Preprocessing and EDA</b> <ul style="list-style-type: none"><li>• Handling Missing Values</li><li>• Data Cleaning</li><li>• Feature Selection</li><li>• Train-Test Split</li><li>• Visualization Using Matplotlib</li></ul> |
| 3      | <b>Regression Techniques</b> <ul style="list-style-type: none"><li>• Implementing Simple Linear Regression</li><li>• Prediction</li><li>• Calculating MAE, MSE, RMSE</li></ul>                                                      |
| 4      | <b>Classification Techniques</b> <ul style="list-style-type: none"><li>• Logistic Regression, KNN, Naïve Bayes</li><li>• Confusion Matrix</li><li>• Accuracy, Precision, Recall</li></ul>                                           |
| 5      | <b>Decision Tree and Clustering</b> <ul style="list-style-type: none"><li>• Building A Decision Tree Model</li><li>• Implementing K-Means Clustering</li><li>• Visualization And Interpretation</li></ul>                           |
| 6      | <b>Model Evaluation and Mini Project</b> <ul style="list-style-type: none"><li>• Cross-Validation (<i>Basic Idea</i>)</li><li>• Overfitting And Underfitting</li><li>• Applying Full ML Workflow on A Small Dataset</li></ul>       |

## Suggested Reading:

1. **Machine Learning using Python** — Manaranjan Pradhan & U. Dinesh Kumar, Publisher: Wiley India (2019; also 2nd edition 2025)
2. **Machine Learning with Python** — Abhishek Vijayvargia, Publisher: BPB Publications (2018)
3. **Applied Machine Learning Solutions with Python** — Siddhanta Bhatta, Publisher: BPB Publications (2022)
4. **Introduction to Machine Learning with Python: A Guide for Data Scientists** — Andreas C. Müller & Sarah Guido (Greyscale Indian Edition), Publisher: Shroff Publishers (2016)
5. **Data Science and Machine Learning Using Python** — Reema Thareja, Publisher: McGraw Hill India

# Research Methodology: BCAC702

| MODULE | TOPIC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | MARKS |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1      | <b>Foundations of Research</b> <ul style="list-style-type: none"> <li>• <b>Meaning, Objectives, And Types of Research:</b> <ul style="list-style-type: none"> <li>○ Basic,</li> <li>○ Applied,</li> <li>○ Exploratory, Descriptive</li> </ul> </li> <li>• Research Process and Characteristics of Good Research</li> <li>• Identifying And Defining a Research Problem</li> <li>• Formulation Of Research Questions and Objectives</li> <li>• Variables In Research and Research Motivation</li> </ul> | 15    |
| 2      | <b>Literature Review and Research Design:</b> <ul style="list-style-type: none"> <li>• Importance Of Literature Review</li> <li>• <b>Sources Of Literature:</b> Journals, Conference Papers, Reports, Online Databases</li> <li>• Research Gap Identification</li> <li>• <b>Types Of Research Design:</b> <ul style="list-style-type: none"> <li>○ Experimental,</li> <li>○ Descriptive,</li> <li>○ Analytical</li> </ul> </li> <li>• Developing Hypotheses and Conceptual Framework</li> </ul>        | 15    |
| 3      | <b>Sampling Techniques and Data Collection Methods:</b> <ul style="list-style-type: none"> <li>• Census Vs Sampling</li> <li>• Probability And Non-Probability Sampling Methods</li> <li>• Primary And Secondary Data Collection</li> <li>• <b>Methods:</b> Surveys, Observation, Interview, Questionnaire</li> <li>• Designing Questionnaires and Schedules</li> </ul>                                                                                                                                | 20    |
| 4      | <b>Statistical Tools for Research:</b> <ul style="list-style-type: none"> <li>• Classification And Tabulation of Data</li> <li>• Measures Of Central Tendency and Dispersion</li> <li>• Correlation And Regression Basics</li> <li>• Estimation Of Population Parameters</li> <li>• <b>Hypothesis testing:</b> t-test, chi-square test, ANOVA-concept</li> </ul>                                                                                                                                       | 20    |
| 5      | <b>Data Analysis, Interpretation and Presentation</b> <ul style="list-style-type: none"> <li>• Data Coding and Data Processing</li> <li>• Use Of Charts, Tables, And Graphs for Presentation</li> <li>• Interpretation Of Statistical Results</li> <li>• Drawing Conclusions and Recommendations</li> <li>• Use Of Software Tools for Analysis (Overview)</li> </ul>                                                                                                                                   | 15    |
| 6      | <b>Research Reporting and Ethics</b> <ul style="list-style-type: none"> <li>• Structure Of Research Report / Thesis / Paper</li> <li>• Writing Abstract, Introduction, Methodology, Results, Conclusion</li> <li>• <b>Referencing styles:</b> (APA/MLA/IEEE)</li> <li>• Plagiarism And Research Ethics</li> <li>• Preparing Research Proposal and Presentation</li> </ul>                                                                                                                              | 15    |

# Research Methodology Lab: BCAC792

| EXP NO. | EXPERIMENTS                                                                                                                                                                                                                  |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | <b>Introduction to Research and Identification of Research Problem:</b> <ul style="list-style-type: none"> <li>Understand different types of research and formulate research problems</li> </ul>                             |
| 2       | <b>Literature Review using Digital Libraries:</b> <ul style="list-style-type: none"> <li>Search and review research papers using databases such as Google Scholar, IEEE Xplore, Springer, and ACM Digital Library</li> </ul> |
| 3       | <b>Research Objectives, Questions and Hypothesis Formulation:</b> <ul style="list-style-type: none"> <li>Develop research objectives, research questions, and hypotheses</li> </ul>                                          |
| 4       | <b>Research Design and Sampling Techniques:</b> <ul style="list-style-type: none"> <li>Learn qualitative, quantitative, and mixed research methods along with sampling techniques</li> </ul>                                 |
| 5       | <b>Designing Questionnaire using Google Forms:</b> <ul style="list-style-type: none"> <li>Create structured questionnaires for data collection</li> </ul>                                                                    |
| 6       | <b>Data Collection and Data Cleaning</b> <ul style="list-style-type: none"> <li>Collect primary/secondary data and perform preprocessing</li> </ul>                                                                          |
| 7       | <b>Statistical Analysis using MS Excel/SPSS/R</b> <ul style="list-style-type: none"> <li>Compute descriptive statistics such as mean, median, mode, standard deviation, and variance</li> </ul>                              |
| 8       | <b>Data Visualization</b> <ul style="list-style-type: none"> <li>Generate charts, graphs, and dashboards for data interpretation</li> </ul>                                                                                  |
| 9       | <b>Hypothesis Testing</b> <ul style="list-style-type: none"> <li>Perform t-test, chi-square test, ANOVA, or correlation analysis based on sample data</li> </ul>                                                             |
| 10      | <b>Citation and Reference Management</b> <ul style="list-style-type: none"> <li>Use tools such as Zotero or Mendeley for citations and bibliography in IEEE/APA formats</li> </ul>                                           |
| 11      | <b>Plagiarism Checking and Ethical Research Practices</b> <ul style="list-style-type: none"> <li>Check similarity using plagiarism detection tools and understand research ethics</li> </ul>                                 |
| 12      | <b>Research Paper Writing using IEEE/APA Format</b> <ul style="list-style-type: none"> <li>Prepare a complete research paper including Abstract, Introduction, Methodology, Results, Conclusion, and References</li> </ul>   |

## Reference Books:

1. **Research Methodology: Methods and Techniques** — C. R. Kothari, Publisher: New Age International Publishers
2. **Research Methodology: A Step-by-Step Guide for Beginners** — Ranjit Kumar, Publisher: Sage Publications India
3. **Research Methods for Business: A Skill-Building Approach** — Uma Sekaran & Roger Bougie, Publisher: Wiley India
4. **Fundamentals of Research Methodology and Statistics** — Yogesh Kumar Singh, Publisher: New Age International Publishers
5. **Business Research Methods** — Donald R. Cooper & Pamela S. Schindler, Publisher: McGraw Hill Education (India)
6. **Social Science Research: Principles, Methods & Practices** — Anol Bhattacharjee, Publisher: University of Florida Press (widely used in India)
7. **Research Design: Qualitative, Quantitative, and Mixed Methods Approaches** — John W. Creswell & J. David Creswell, Publisher: Sage Publications (International edition often available in India)

# Cyber Security: BCAC703

| MODULE | TOPICS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | MARKS |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1      | <b>Networking Fundamentals for Cyber Security</b> <ul style="list-style-type: none"> <li>• Introduction to Networking</li> <li>• OSI &amp; TCP/IP Framework</li> <li>• Transmission Media (with Case Study)</li> <li>• Network Devices (with Case Study)</li> <li>• TCP/IP Framework in detail</li> <li>• Subnet Masking (with Case Study)</li> <li>• Three-way Handshaking</li> <li>• NAT</li> <li>• Different Protocols &amp; their uses</li> </ul>                                                                               | 20    |
| 2      | <b>Cyber Security Foundations</b> <ul style="list-style-type: none"> <li>• Cyber Security Definition</li> <li>• Security Triad (CIA) with Case Study</li> <li>• Hacking &amp; Ethical Hacking</li> <li>• Types of Hackers</li> <li>• Phases of Hacking (<i>Reconnaissance, Scanning, Gaining Access, Maintaining Access, Covering Tracks</i>)</li> <li>• Cryptography &amp; Watermarking and their use</li> <li>• Types of Cryptography</li> <li>• Symmetric vs Asymmetric Algorithms</li> <li>• RSA Algorithm explained</li> </ul> | 20    |
| 3      | <b>Information Security Management</b> <ul style="list-style-type: none"> <li>• Introduction to ISM</li> <li>• Goals of Information Security</li> <li>• ISMS (Information Security Management System) working procedure</li> <li>• Security Policies</li> <li>• Risk Management</li> <li>• Information Security Audit (with Case Study)</li> </ul>                                                                                                                                                                                  | 15    |
| 4      | <b>Cyber Attacks</b> <ul style="list-style-type: none"> <li>• <b>Types of Attacks:</b> Passive &amp; Active Attacks</li> <li>• Malware</li> <li>• Phishing</li> <li>• Denial of Service</li> <li>• SQL Injection</li> <li>• Social Engineering</li> <li>• Insider Threats</li> <li>• Case Studies of Major Cyber Attacks</li> </ul>                                                                                                                                                                                                 | 20    |
| 5      | <b>Technological Aspects of Cyber Security</b> <ul style="list-style-type: none"> <li>• Firewalls</li> <li>• IDS &amp; IPS</li> <li>• VPNs</li> <li>• Security Protocols (<i>SSL, TLS, IPsec</i>)</li> </ul>                                                                                                                                                                                                                                                                                                                        | 25    |



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|  | <ul style="list-style-type: none"> <li>• Intrusion Detection Systems</li> <li>• Honeypots</li> <li>• Multi-Factor Authentication</li> <li>• Emerging Technologies in Cyber Security (<i>AI, Blockchain in Security, Quantum Cryptography</i>)</li> </ul> |  |
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## Books & References:

1. **Fundamentals of Cyber Security (Principles, Theory & Practices)** — Bhushan, M., & Rathore, R. S. (2017), Publisher: BPB Publications, New Delhi
2. **Cyber Security in India: Education, Research and Training** — Shukla, S. K., & Agrawal, M. (2020), Publisher: Springer, Singapore
3. **Cyber Security and Cyber Laws** — Basta, A., Basta, N., Brown, M., & Kumar, R. (2018), Publisher: Cengage Learning India, New Delhi
4. **Cyber Security** — Godbole, N. (2011), Publisher: Wiley India, New Delhi
5. **Cryptography and Network Security** — Mukhopadhyay, D., & Forouzan, B. A. (2011), Publisher: Tata McGraw-Hill Education, New Delhi

# Consumer Behavior: MIM701A

| MODULE | TOPICS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | % of QA |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 1      | <b>Introduction To Consumer Behavior</b> <ul style="list-style-type: none"> <li>• Evolution of consumer behavior</li> <li>• Understanding consumers and market segments</li> <li>• Consumer behavior and marketing strategy</li> <li>• Psychographic dimensions</li> <li>• Consumer motivation</li> <li>• Perception</li> <li>• Personality</li> <li>• Information processing</li> <li>• Attitude formation and attitude change</li> <li>• Scope and their applications</li> <li>• Information search process</li> <li>• Evaluative criteria and decision rules</li> <li>• Building customer satisfaction</li> <li>• 7'Os of consumer behavior</li> </ul> | 10%     |
| 2      | <b>Factors Affecting Consumer Behavior</b> <ul style="list-style-type: none"> <li>• Factors influencing consumer behavior</li> <li>• <b>External Influences</b> <ul style="list-style-type: none"> <li>◦ Culture</li> <li>◦ Sub Culture</li> <li>◦ Social Class</li> <li>◦ Reference Groups</li> <li>◦ Family</li> </ul> </li> <li>• <b>Internal Influences</b> <ul style="list-style-type: none"> <li>◦ Needs &amp; Motivations</li> <li>◦ Perception</li> <li>◦ Personality</li> <li>◦ Lifestyle</li> <li>◦ Values</li> <li>◦ Learning</li> <li>◦ Memory</li> <li>◦ Beliefs &amp; Attitudes</li> </ul> </li> </ul>                                      | 10%     |
| 3      | <b>Social and Cultural Environment</b> <ul style="list-style-type: none"> <li>• Economic</li> <li>• Demographic</li> <li>• Cross Cultural and Socio-Cultural Influences</li> <li>• Cultural Relevance to Marketing Decisions</li> <li>• Characteristics of Culture</li> <li>• Cultural Values</li> <li>• Cultural Changes</li> <li>• Cross Cultural Understandings</li> <li>• Social Stratification</li> <li>• Reference Groups and Family Influences</li> <li>• Personal Influence</li> </ul>                                                                                                                                                            | 20%     |

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|   | <ul style="list-style-type: none"> <li>• <b>Family</b> <ul style="list-style-type: none"> <li>○ Role &amp; Structure</li> <li>○ Family Life Cycle</li> <li>○ Purchasing Decisions</li> <li>○ Changing Role of Families</li> </ul> </li> <li>• Role of Reference Group</li> <li>• Types of Reference Group</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| 4 | <p><b>Consumers Need, Motivation, Personality, Perception and Learning</b></p> <ul style="list-style-type: none"> <li>• <b>Motivation</b> <ul style="list-style-type: none"> <li>○ Needs</li> <li>○ Goals</li> <li>○ Motive Arousal</li> <li>○ Maslow Hierarchy of Needs</li> <li>○ Alderfer's ERG Theory</li> <li>○ Herzberg Motivation-Hygiene Theory</li> <li>○ McClelland's Achievement Theory</li> <li>○ McGregor's Theory X and Y</li> <li>○ Vroom's Expectancy Theory</li> <li>○ Porter's Expectancy Theory</li> <li>○ Freud's Theory of Motivation</li> </ul> </li> <li>• <b>Personality</b> <ul style="list-style-type: none"> <li>○ Meaning and Concept of Personality</li> <li>○ Determinant of Personality</li> <li>○ <b>Theory of Personality</b> <ul style="list-style-type: none"> <li>▪ Self-concept Theory</li> <li>▪ Psychoanalytic Theory</li> <li>▪ Neo-Freudian Theory</li> <li>▪ Trait Theory</li> <li>▪ Humanistic Theory</li> <li>▪ Social-cognitive Theory</li> </ul> </li> <li>○ Measurement of Personality</li> <li>○ Development of Personality</li> </ul> </li> <li>• <b>Perception</b> <ul style="list-style-type: none"> <li>○ <b>Definition and Importance</b> <ul style="list-style-type: none"> <li>▪ Defining Perception</li> <li>▪ Nature of Perception</li> <li>▪ Significance in Consumer Behavior</li> </ul> </li> <li>○ <b>The Perceptual Process</b> <ul style="list-style-type: none"> <li>▪ Exposure</li> <li>▪ Attention</li> <li>▪ Organization</li> <li>▪ Interpretation</li> <li>▪ Retention</li> </ul> </li> <li>○ <b>Elements of Perception</b> <ul style="list-style-type: none"> <li>▪ Sensation</li> <li>▪ Threshold</li> <li>▪ Perceptual Biases</li> </ul> </li> <li>○ <b>Sensory Dynamics</b> <ul style="list-style-type: none"> <li>▪ Sight</li> <li>▪ Sound</li> </ul> </li> </ul> </li> </ul> | 20% |

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|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
|   | <ul style="list-style-type: none"> <li>▪ Touch</li> <li>▪ Smell</li> <li>▪ Taste</li> <li>○ <b>Barriers to Accurate Perception</b> <ul style="list-style-type: none"> <li>▪ Selective Attention</li> <li>▪ Perceptual Defense</li> </ul> </li> <li>○ <b>Perception and Consumer Decision-Making</b> <ul style="list-style-type: none"> <li>▪ Perception of Products and Brands</li> <li>▪ Perception of Price</li> <li>▪ Perception of Risk</li> <li>▪ Perception and Advertising</li> </ul> </li> <li>○ Perceptual Mapping</li> <li>○ Subliminal Perception</li> <li>○ Consumer Imagery</li> <li>• <b>Learning</b> <ul style="list-style-type: none"> <li>○ Meaning</li> <li>○ Definition</li> <li>○ Concept</li> <li>○ Relationship with Consumer Behavior</li> <li>○ Theories of Learning <ul style="list-style-type: none"> <li>▪ <b>Behavioral Learning Theory</b> <ul style="list-style-type: none"> <li>▪ Theory of Classical Conditioning (Stimulus-Response)</li> <li>▪ Cognitive Associative Learning (Memory/Trial-Error Based)</li> <li>▪ Operant or Instrumental Conditioning (Result/Reinforcement Based)</li> <li>▪ Social Learning Theory</li> </ul> </li> <li>▪ <b>Cognitive Theories</b> <ul style="list-style-type: none"> <li>▪ Social Cognitive Theory</li> <li>▪ Cognitive Behavioral Theory</li> <li>▪ Constructivism</li> </ul> </li> </ul> </li> </ul> </li> </ul> |     |
| 5 | <p><b>Consumer Decision Making Process</b></p> <ul style="list-style-type: none"> <li>• Types of Consumer Decisions</li> <li>• <b>Consumer Decision Making Process</b> <ul style="list-style-type: none"> <li>○ Problem Recognition</li> <li>○ Information Search</li> <li>○ Alternative Evaluation</li> <li>○ Purchase Selection</li> <li>○ Post Purchase Evaluation</li> </ul> </li> <li>• Buying Pattern in the New Digital Era</li> <li>• <b>Four Views of Consumer Decision Rules</b> <ul style="list-style-type: none"> <li>○ Economic Man</li> <li>○ Passive Man</li> <li>○ Emotional Man</li> <li>○ Cognitive Man</li> </ul> </li> <li>• <b>Models of Consumer Decision Making</b> <ul style="list-style-type: none"> <li>○ Nicosia Model</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 20% |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |
| 6 | <b>Marketing Communications, Decision Making Models, Consumer Rights</b> <ul style="list-style-type: none"> <li>• Marketing Communication Process</li> <li>• <b>Types of Communication Systems</b> <ul style="list-style-type: none"> <li>◦ Interpersonal</li> <li>◦ Impersonal</li> <li>◦ Persuasive Communication</li> </ul> </li> <li>• <b>Consumer Decision Making Models</b> <ul style="list-style-type: none"> <li>◦ Black Box Model</li> <li>◦ Economic Model</li> <li>◦ Howard Model</li> <li>◦ Howard-Sheth Model</li> <li>◦ EKB Model</li> <li>◦ Webster and Wind Model</li> <li>◦ Sheth Industrial Buyer Behaviour Model</li> </ul> </li> <li>• Consumer Protection Act 1986</li> <li>• Rights of Consumers</li> </ul> | 20% |

### Reading Suggestions:

- **Consumer Behaviour** – Anis Chattopadhyay – *Taurean Publications*
- **Consumer Behavior** – Kumar Leon G., Schiffman, Joe Wisenblit, S. Ramesh – *Pearson*
- **Consumer Behavior** – Schiffman – *Pearson*

# Strategic Management: MIM702A

| MODULE | TOPICS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | % of QA |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 1      | <b>Strategic Management: An Overview of Strategic Management</b> <ul style="list-style-type: none"> <li>• Understanding Strategy</li> <li>• Scope and Importance of Strategies</li> <li>• Mission</li> <li>• Vision</li> <li>• Goals</li> <li>• Objectives</li> <li>• Defining and Explaining Strategy</li> <li>• Levels at Which Strategy Operates</li> <li>• Strategic Decision Making</li> <li>• The Process of Strategic Management</li> </ul>                                                                               | 15%     |
| 2      | <b>Strategy Formulation</b> <ul style="list-style-type: none"> <li>• Environmental Scanning</li> <li>• SWOT Analysis</li> <li>• Internal and External Environmental Analysis</li> <li>• <b>Competition Analysis</b> <ul style="list-style-type: none"> <li>◦ Porter's Five Forces Theory</li> </ul> </li> <li>• Generic Strategies</li> <li>• Competitive Advantage</li> <li>• Value Chain Analysis</li> <li>• McKinsey's 7S Framework</li> <li>• Balanced Scorecard</li> <li>• BCG Matrix</li> <li>• PESTLE Analysis</li> </ul> | 20%     |
| 3      | <b>Strategy Implementation</b> <ul style="list-style-type: none"> <li>• Organisational Structure</li> <li>• Analysing Strategic Change</li> <li>• Managing Strategic Change</li> <li>• Issues in Strategy Implementation</li> <li>• <b>Behavioural Issues</b> <ul style="list-style-type: none"> <li>◦ Leadership</li> <li>◦ Corporate Culture</li> <li>◦ Social Responsibilities</li> <li>◦ Ethics</li> </ul> </li> </ul>                                                                                                       | 15%     |
| 4      | <b>Strategy Evaluation and Control</b> <ul style="list-style-type: none"> <li>• Designing Strategic Control System</li> <li>• <b>Functional Issues</b> <ul style="list-style-type: none"> <li>◦ Operational/Production</li> <li>◦ Marketing</li> <li>◦ Financial</li> <li>◦ Human Resource Management</li> <li>◦ Environmental Factors</li> </ul> </li> <li>• Information for Strategic Control</li> <li>• Techniques of Strategic Evaluation and Control</li> <li>• Implementing Strategic Control</li> </ul>                   | 25%     |

|   |                                                                                                                                                                                                                                                                                                                                                                                                          |     |
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| 5 | <b>Business Policies</b> <ul style="list-style-type: none"> <li>• Introduction</li> <li>• Overview of Business Policies</li> <li>• Importance of Business Policies</li> <li>• Definition of Business Policy</li> <li>• Procedure</li> <li>• Process</li> <li>• Types of Policies</li> <li>• Factors Considered Before Framing Policies</li> <li>• Steps Involved in Framing Business Policies</li> </ul> | 25% |
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### Reading Suggestion:

1. **Strategic Management** – Anis Chattopadhyay – Taurean Publications
2. **Business Policy and Strategic Management** – P. Subba Rao – Himalaya Publishing House
3. **Business Policy and Strategic Management** (2nd ED) – Azhar Kazmi – McGraw-Hill